

Session 2 - Forecasting & Valuation Inputs

Applied Exercises with Solutions

Exercise 1 - MedNova Technologies: Forecasting Revenue from Business Drivers

CASE

MedNova Technologies is a European medical-device company. In 2026, it generated revenue of EUR 2,400 million. Its Diagnostic Devices division accounted for 60% of group revenue, while Digital Health accounted for the remaining 40%.

The market for Diagnostic Devices is expected to grow by 2.5% in 2027. Because MedNova recently strengthened its hospital distribution network, analysts expect market-share improvements to contribute an additional 0.5 percentage points to the company's revenue growth in this segment.

The Digital Health market is expected to expand by 9%. However, several competitors are introducing similar products, and analysts expect competitive pressure to reduce MedNova's growth by approximately 1 percentage point relative to the market. Management has communicated that it expects group revenue growth to remain in the "mid-single-digit" range.

REQUIRED

Develop a defensible 2027 revenue forecast for MedNova. Forecast the two segments separately before calculating consolidated revenue growth. Explain whether your estimate is consistent with management guidance and identify the information you would investigate before using these assumptions in a DCF. Also describe how the economic drivers might change in a downside and an upside scenario.

Exercise 2 - Aurelia Foods: Historical Growth Is Not the Forecast

CASE

Aurelia Foods is a branded consumer-products company. Revenue increased from EUR 3.8 billion in 2022 to EUR 5.0 billion in 2026.

This strong growth was partly driven by substantial price increases during the inflationary period. Aurelia also acquired a regional food brand during 2024. Management expects inflation to normalize and has not announced another major acquisition.

Independent industry forecasts indicate that Aurelia's categories are likely to grow by approximately 3.5% per year over the next several years. Management believes new product launches and improved distribution could contribute approximately another 0.5 percentage points to organic company growth.

REQUIRED

Assess whether Aurelia's historical revenue growth provides an appropriate forecast for the future. Estimate the historical CAGR and develop a more defensible forward-looking revenue assumption. Explain why the forecast may reasonably differ from historical performance.

Exercise 3 - BelleVue Luxury: Building a Margin Forecast

CASE

BelleVue Luxury generated revenue of EUR 8 billion in 2026 and reported an EBIT margin of 20%.

For 2027, management expects higher selling prices and a greater contribution from premium products. Procurement contracts are also expected to lower logistics and sourcing costs. In addition, revenue growth should improve fixed-cost absorption. At the same time, BelleVue plans to increase advertising and brand-development expenditure.

Analysts estimate the following effects on operating margin: pricing and product mix +0.6 percentage points; lower input and logistics costs +0.2 percentage points; operating leverage +0.4 percentage points; additional brand investment -0.5 percentage points. Revenue is forecast at EUR 8.6 billion.

REQUIRED

Estimate BelleVue's 2027 EBIT margin and operating profit. Then assess whether the same margin improvement should automatically be extrapolated into later years.

Exercise 4 - GreenCore Industries: Growth, ROIC and Reinvestment

CASE

GreenCore Industries currently generates NOPAT of EUR 300 million and earns a forecast ROIC of approximately 15%. Management believes NOPAT can grow by around 6% annually over the medium term.

The valuation model currently contains very little reinvestment because CapEx is approximately equal to depreciation and working capital is assumed to remain unchanged.

REQUIRED

Assess whether the operating assumptions are economically consistent. Using forecast ROIC as an approximation for the return on incremental investment, estimate the reinvestment rate implied by the 6% growth assumption and the amount of annual reinvestment required.

Exercise 5 - Ferro Manufacturing: Two Ways to Forecast Reinvestment

CASE

Ferro Manufacturing forecasts the following for 2027: NOPAT EUR 500m, CapEx EUR 260m, D&A EUR 190m, and an increase in operating NWC of EUR 30m.

Management expects operating profit to grow by approximately 5% annually. Forecast ROIC is approximately 12%.

REQUIRED

Compare the reinvestment implied by the detailed operating forecast with the reinvestment suggested by the fundamental growth/return relationship. Explain what the difference tells you about the model.

Exercise 6 - Nexora Software: Capitalizing R&D

CASE

Nexora Software develops enterprise cybersecurity software. In 2026 it reports EBIT of EUR 280m, current R&D expense of EUR 120m, estimated amortization of past R&D of EUR 75m, a beginning capitalized R&D asset of EUR 240m, and a tax rate of 25%.

Because Nexora's R&D projects create products and technology that generate benefits over several years, the analyst decides to treat R&D as an investment for valuation purposes.

REQUIRED

Recast Nexora's operating performance after capitalizing R&D. Calculate adjusted EBIT, adjusted NOPAT and the closing R&D asset. Explain the economic logic behind the adjustments.

Exercise 7 - Nexora Software: R&D and FCFF

CASE

Continue with Nexora Software. In addition to the information above, the company reports D&A of EUR 90m, CapEx of EUR 130m, and an increase in operating NWC of EUR 20m. Reported NOPAT before the R&D capitalization adjustment is EUR 210m.

REQUIRED

Calculate FCFF under the standard accounting treatment and after treating R&D as an investment. Explain why R&D capitalization does not create additional cash.

Exercise 8 - NordMart Retail: Working Capital and FCFF

CASE

NordMart operates supermarkets across Europe. Its forecasts show revenue of EUR 6,000m in 2026 and EUR 6,600m in 2027. Operating NWC is negative EUR 480m in 2026 and is forecast to fall further to negative EUR 594m in 2027.

Management explains that customers pay immediately while suppliers provide payment terms. NordMart has also negotiated longer supplier payment periods for 2027.

REQUIRED

Evaluate the effect of the forecast working-capital change on FCFF and assess whether the benefit can reasonably be assumed to continue indefinitely.

- A favorable working-capital release can boost FCFF today without necessarily creating a sustainable long-run source of value.

Exercise 9 - Titan Components: Maintenance and Growth CapEx

CASE

Titan Components supplies parts for electric vehicles. The company reports revenue growth of 7%, capacity utilization of 94%, D&A of EUR 180m and CapEx of EUR 150m.

Management has announced a new plant requiring EUR 250m of investment over the next two years. The facility is needed to support expected growth in customer orders. A junior analyst proposes forecasting CapEx at EUR 150m annually because that is the most recent reported amount.

REQUIRED

Assess whether the analyst's forecast is appropriate. Explain how maintenance CapEx and growth CapEx should be reflected in Titan's valuation.

Exercise 10 - StreamWave Media: Customer and ARPU Forecasting

CASE

StreamWave Media operates a subscription-based streaming platform. At the end of 2026 it has 20 million subscribers and annual ARPU of EUR 120.

Management expects subscriber numbers to increase by 8% in 2027. However, StreamWave plans to introduce a lower-priced subscription package, causing average ARPU to fall by 2%. Management has guided investors toward group revenue growth of approximately 6%.

REQUIRED

Use StreamWave's underlying business drivers to estimate 2027 revenue and evaluate management's guidance.

Exercise 11 - Helio Energy: Forecast Horizon and Terminal Value

CASE

Helio Energy is expanding a portfolio of renewable-energy projects. At the end of Year 5 of the analyst's DCF, revenue is still growing at 8%, operating margins continue to improve, ROIC is approximately 17%, and several large capacity projects remain under construction.

The analyst nevertheless intends to calculate terminal value immediately after Year 5 because the firm's standard valuation template uses a five-year explicit forecast.

REQUIRED

Assess whether Year 5 represents an appropriate terminal year. Explain what should happen to the company's operating assumptions before a stable-growth terminal value is applied.

Exercise 12 - Arcturus Industrial: Bottom-Up Beta and WACC

CASE

Arcturus Industrial manufactures specialized industrial equipment. Three comparable firms have the following characteristics.

REQUIRED

Estimate Arcturus's bottom-up beta, cost of equity and WACC. Assume a tax rate of 25%, a target D/E ratio of 35%, a risk-free rate of 2.5%, an ERP of 5.0%, and a pre-tax cost of debt of 4.5%.

Exercise 13 - Atlas Global Services: Risk-Free Rate and Country Risk

CASE

Atlas Global Services is headquartered in France, but its operations are internationally diversified. Approximately 50% of operating exposure is in the Eurozone, 30% in the United States and 20% in emerging markets.

The valuation is being performed in euros. Assume a euro risk-free rate of 2.4%, a US Treasury yield of 3.8%, an ERP of 5.0%, a beta of 1.10, and a country-risk premium of 3.0% for the relevant emerging-market exposure. For this exercise, operating revenue exposure is used as a simplified proxy for country-risk exposure.

REQUIRED

Determine the appropriate risk-free rate and estimate Atlas's cost of equity, including an approximate country-risk adjustment.

Exercise 14 - Solstice Hotels: Build vs Buy

CASE

Solstice Hotels generated revenue of EUR 1.8 billion in 2026. During the previous four years, reported revenue growth averaged approximately 7% annually. However, part of this growth came from acquiring regional hotel chains.

Management currently plans to open 20 new hotels organically during the next three years but has not announced additional acquisitions. Industry demand is expected to grow approximately 3%. A junior analyst proposes forecasting 7% organic revenue growth because this matches Solstice's historical growth rate.

REQUIRED

Assess the analyst's approach and develop a more appropriate framework for forecasting Solstice's future revenue.